

CYBERSECURITY-LABS

INFORMATION SECURITY

LAB REPORT



Práctica: Automatización con Ansible

Informe de práctica

Víctor José Santos Guzmán

Information Security

Cybersecurity / Cloud / Networks / GRC



Paso 1: [Actualización de paquetes en el clúster] Inicié el mantenimiento conectándome a los tres servidores simultáneamente para asegurar que todos los nodos estuvieran sincronizados. Ejecuté el comando combinado `sudo apt-get upgrade && sudo apt-get update` para refrescar las listas de repositorios y aplicar las actualizaciones disponibles, garantizando así que el sistema base estuviera al día antes de iniciar configuraciones específicas.

```
victor_2025064@victor-seven:~$ sudo apt-get upgrade && sudo apt-get update
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [124 kB]
Get:2 http://do.archive.ubuntu.com/ubuntu noble InRelease [124 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,317 kB]
Get:4 http://do.archive.ubuntu.com/ubuntu noble-backports InRelease [124 kB]
Get:5 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,468 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,317 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [214 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 C-n-f Metadata [9,452 B]
Get:10 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1,153 kB]
Get:11 http://do.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [303 kB]
Get:12 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [178 kB]
Get:13 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 C-n-f Metadata [15.7 kB]
Get:14 http://do.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2,307 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [1690 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:17 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [909 kB]
Get:18 http://do.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [352 kB]
Get:19 http://do.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:20 http://do.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,500 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [200 kB]
Get:22 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.3 kB]
Get:23 http://security.ubuntu.com/ubuntu noble-security/universe amd64 C-n-f Metadata [19.4 kB]
Get:24 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [5,566 B]
Get:25 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:26 http://do.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [804 kB]
Get:27 http://do.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [379 kB]
Get:28 http://do.archive.ubuntu.com/ubuntu noble-updates/universe amd64 C-n-f Metadata [31.4 kB]
Get:29 http://do.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [5,809 B]
Get:30 http://do.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:31 http://do.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,448 B]
Get:32 http://do.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:33 http://do.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [11.0 kB]
Get:34 http://do.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 13.0 MB in 4s (3,115 kB/s)
sudo apt-get upgrade && sudo apt-get update
Reading package lists... Done
victor_2025064@victor-seven:~$
```

Paso 2: [Configuración del Firewall UFW] Procedí a asegurar la red configurando el firewall en todos los servidores para controlar el tráfico entrante. Habilité las conexiones necesarias ejecutando `sudo ufw allow OpenSSH` y `sudo ufw allow 80/tcp`, activé el servicio con `sudo ufw enable` y verifiqué que las reglas se aplicaran correctamente en IPv4 e IPv6 mediante el comando `sudo ufw status`.

```
victor_2025064@victor-seven:~$ sudo su
root@victor-server:/home/victor_2025064# sudo ufw allow OpenSSH
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
root@victor-server:/home/victor_2025064# sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y/n) ? y
Firewall is active and enabled on system startup
root@victor-server:/home/victor_2025064# sudo ufw status
Status: active

To Action From
--
OpenSSH ALLOW Anywhere
80/tcp ALLOW Anywhere (v6)
OpenSSH (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)

root@victor-server:/home/victor_2025064#
```

```
server2025064@server2:~$ sudo ufw allow OpenSSH
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Command may disrupt existing ssh connections. Proceed with operation (y/n) ? y
Firewall is active and enabled on system startup
Status: active

To Action From
--
OpenSSH ALLOW Anywhere
80/tcp ALLOW Anywhere (v6)
OpenSSH (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)

server2025064@server2:~$
```

```
server3025064@server3:~$ sudo ufw allow OpenSSH
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Command may disrupt existing ssh connections. Proceed with operation (y/n) ? y
Firewall is active and enabled on system startup
Status: active

To Action From
--
OpenSSH ALLOW Anywhere
80/tcp ALLOW Anywhere (v6)
OpenSSH (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)

server3025064@server3:~$
```

Paso 3: [Creación e indexación del repositorio local] Preparé un repositorio local creando el directorio con `mkdir -p /repo` y descargué el instalador de Nginx usando `sudo apt download nginx`. Para hacer el repositorio funcional, generé el archivo de índice comprimido ejecutando el comando `sudo dpkg-scanpackages . /dev/null | gzip -9c | sudo tee packages.gz`, omitiendo errores previos de permisos al usar `sudo` correctamente.

```
root@victor-server:/home/victor_20250664# mkdir -p /repo
root@victor-server:/home/victor_20250664# cd /repo
root@victor-server:/repo# sudo apt download nginx
Get:1 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.5 [520 kB]
Fetched 520 kB in 1s (823 kB/s)
W: Download is performed unsandboxed as root as file '/repo/nginx_1.24.0-2ubuntu7.5_amd64.deb' couldn't be accessed b
y user '_apt'. - pkgAcquire::Run (13: Permission denied)
root@victor-server:/repo# cd /repo
sudo apt download nginx
sudo cp nginx *.deb /repo/
cp: 'nginx_1.24.0-2ubuntu7.5_amd64.deb' and '/repo/nginx_1.24.0-2ubuntu7.5_amd64.deb' are the same file.
root@victor-server:/repo# sudo dpkg-scanpackages . /dev/null | gzip -9c | sudo tee packages.gz
```

Paso 4: [Validación del contenido del repositorio] Finalmente, realicé una comprobación del directorio recién creado para confirmar la integridad de los archivos generados. Utilicé el comando `ls -l /repo` para listar el contenido, verificando exitosamente que tanto el paquete `.deb` de Nginx como el archivo de índice `packages.gz` existen y tienen los permisos de lectura adecuados para ser utilizados por el sistema.

```
root@victor-server:/repo# ls -l /repo
total 512
-rw-r--r-- 1 root root 520182 Aug 25 14:28 nginx_1.24.0-2ubuntu7.5_amd64.de
-rw-r--r-- 1 root root 694 Nov 24 00:43 packages.gz
root@victor-server:/repo#
```

Paso 5: [Inicio de instalación de Apache] Procedí a instalar el servidor web necesario para exponer el repositorio ejecutando el comando `sudo apt install apache2`. Al recibir el resumen de las dependencias y el espacio en disco requerido, confirmé la operación ingresando `y`, lo que inició la descarga de los binarios principales y las librerías de soporte necesarias para el funcionamiento del servicio HTTP.

```
root@victor-server:/repo# sudo apt install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
  liblua5.4-0 ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
  liblua5.4-0 ssl-cert
0 upgraded, 10 newly installed, 0 to remove and 4 not upgraded.
Need to get 2,086 kB of archives.
After this operation, 8,090 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]
Get:2 http://do.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7 [91.9 kB]
Get:3 http://do.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7 [11.2 kB]
Get:4 http://do.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7 [9,116 B]
Get:5 http://do.archive.ubuntu.com/ubuntu noble/main amd64 liblua5.4-0 amd64 5.4.6-3build2 [166 kB]
Get:6 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd64 2.4.58-1ubuntu8.8 [1,331 kB]
Get:7 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu8.8 [163 kB]
86% [Working]
```

Paso 6: [Finalización del despliegue de paquetes] Supervisé la fase de desempaquetado y configuración automática donde el gestor apt instaló los archivos en el sistema. Durante este proceso, verifiqué visualmente que se procesaran los disparadores (triggers), se crearan los enlaces simbólicos para systemd y que la instalación concluyera exitosamente sin errores de dependencias, dejando el software listo para ser configurado.

```
root@victor-server:/repo# apt install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
  liblua5.4-0 ssl-cert
Suggested packages:
  apache2-doc apache2-ssl-privkey | apache2-ssl-certificate www-browser
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
  liblua5.4-0 ssl-cert
0 upgraded, 10 newly installed, 0 to remove and 4 not upgraded.
Need to get 2,064 kB of archives.
After this operation, 9,500 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://de.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]
Get:2 http://de.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7 [91.9 kB]
Get:3 http://de.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7 [11.2 kB]
Get:4 http://de.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7 [9,116 B]
Get:5 http://de.archive.ubuntu.com/ubuntu noble/main amd64 liblua5.4-0 amd64 5.4.6-2build2 [166 kB]
Get:6 http://de.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd64 2.4.58-1ubuntu0.8 [1,331 kB]
Get:7 http://de.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu0.8 [163 kB]
Get:8 http://de.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-utils amd64 2.4.58-1ubuntu0.8 [97.7 kB]
Get:9 http://de.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2 amd64 2.4.58-1ubuntu0.8 [50.2 kB]
Get:10 http://de.archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]
Fetched 2,064 kB in 1s (1,835 kB/s)
Preconfiguring packages...
Selecting previously unselected package libapr1t64:amd64.
(Reading database ... 17460 files and directories currently installed.)
Preparing to unpack .../0-libapr1t64_1.7.2-3.1ubuntu0.1_amd64.deb ...
Unpacking libapr1t64:amd64 (1.7.2-3.1ubuntu0.1) ...
Selecting previously unselected package libaprutil1t64:amd64.
Preparing to unpack .../1-libaprutil1t64_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1t64:amd64 (1.6.3-1.1ubuntu7) ...
Selecting previously unselected package libaprutil1-dbd-sqlite3:amd64.
Preparing to unpack .../2-libaprutil1-dbd-sqlite3_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1-dbd-sqlite3:amd64 (1.6.3-1.1ubuntu7) ...
Selecting previously unselected package libaprutil1-ldap:amd64.
Preparing to unpack .../3-libaprutil1-ldap_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1-ldap:amd64 (1.6.3-1.1ubuntu7) ...
Selecting previously unselected package liblua5.4-0:amd64.
Preparing to unpack .../4-liblua5.4-0_5.4.6-2build2_amd64.deb ...
Unpacking liblua5.4-0:amd64 (5.4.6-2build2) ...
Selecting previously unselected package apache2-bin.
Preparing to unpack .../5-apache2-bin_2.4.58-1ubuntu0.8_amd64.deb ...
Unpacking apache2-bin (2.4.58-1ubuntu0.8) ...
Selecting previously unselected package apache2-data.
Preparing to unpack .../6-apache2-data_2.4.58-1ubuntu0.8_all.deb ...
Unpacking apache2-data (2.4.58-1ubuntu0.8) ...
Selecting previously unselected package apache2-utils.
Preparing to unpack .../7-apache2-utils_2.4.58-1ubuntu0.8_amd64.deb ...
Unpacking apache2-utils (2.4.58-1ubuntu0.8) ...
Selecting previously unselected package apache2.
Preparing to unpack .../8-apache2_2.4.58-1ubuntu0.8_amd64.deb ...
Unpacking apache2 (2.4.58-1ubuntu0.8) ...
Preparing to unpack .../9-ssl-cert_1.1.2ubuntu1_all.deb ...
Unpacking ssl-cert (1.1.2ubuntu1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/ssl-cert.service → /usr/lib/systemd/system/ssl-cert.servi
root@victor-server:/repo#
```

Paso 7: [Creación del Virtual Host para el repositorio] Configuré el acceso web al directorio del repositorio creando un archivo de configuración dedicado. Utilicé una tubería con echo y sudo tee /etc/apache2/sites-available/repo.conf para definir un bloque <Directory> y un Alias, estableciendo las opciones "Indexes" y "FollowSymLinks", y otorgando permisos de acceso global para asegurar que los clientes puedan listar y descargar los paquetes.

```
root@victor-server:/repo#
root@victor-server:/repo# echo "Alias /repo /repo
<Directory /repo>
    Options Indexes FollowSymLinks
    AllowOverride None
    Require all granted
</Directory>" | sudo tee /etc/apache2/sites-available/repo.conf
Alias /repo /repo
<Directory /repo>
    Options Indexes FollowSymLinks
    AllowOverride None
    Require all granted
</Directory>
root@victor-server:/repo#
```

Paso 8: [Modificación del puerto de escucha] Para evitar conflictos con el puerto estándar, cambié el puerto de escucha del servidor web al 8080. Modifiqué la configuración del sitio usando sed y actualicé la configuración global de puertos ejecutando el comando echo "Listen 8080" | sudo tee /etc/apache2/ports.conf, instruyendo así a Apache para que acepte conexiones entrantes únicamente a través de este puerto específico.

```
root@victor-server:/repo# sudo sed -i 's/80/8080/g' /etc/apache2/sites-available/repo.conf
root@victor-server:/repo# sudo echo "Listen 8080" | sudo tee /etc/apache2/ports.conf
Listen 8080
root@victor-server:/repo#
```

Paso 9: [Arranque y verificación del servicio] Para aplicar las nuevas configuraciones, habilité el arranque automático y reinicié el servicio ejecutando `sudo systemctl enable apache2 --now` seguido de `sudo systemctl restart apache2`. Finalmente, validé la operatividad del servidor mediante `sudo systemctl status apache2`, confirmando que el estado es "active (running)" y que los procesos están funcionando correctamente

```
root@victor-server:/repo# sudo systemctl enable apache2 --now
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
root@victor-server:/repo# sudo systemctl restart apache2
root@victor-server:/repo# sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2025-11-24 00:47:55 UTC; 5s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 22725 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 22728 (apache2)
    Tasks: 55 (limit: 4548)
   Memory: 5.1M (peak: 5.2M)
      CPU: 18ms
   CGroup: /system.slice/apache2.service
           └─22728 /usr/sbin/apache2 -k start
             └─22730 /usr/sbin/apache2 -k start
               └─22731 /usr/sbin/apache2 -k start

Nov 24 00:47:55 victor-server systemd[1]: Starting apache2.service - The Apache HTTP Server...
Nov 24 00:47:55 victor-server apachectl[22727]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name: [127.0.0.1]
Nov 24 00:47:55 victor-server systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-17/17 (END)
root@victor-server:/repo#
```

Paso 10: [Validación HTTP del Repositorio] Para confirmar que el servidor web Apache está sirviendo el contenido correctamente, realicé una petición local utilizando el comando `curl http://localhost:8080/repo/`. Analicé la salida HTML cruda en la terminal, verificando que la estructura de directorios, los encabezados y los archivos del repositorio son visibles y accesibles a través del puerto configurado.

```
root@victor-server:/repo# curl http://localhost:8080/repo/
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 3.2 Final//EN">
<html>
<head>
<title>Index of /repo</title>
</head>
<body>
<h1>Index of /repo</h1>
<table>
<tr><th valign="top"></th><th><a href="?C=N;O=D">Name</a></th><th><a href="?C=M;O=A">Last modified</a></th><th><a href="?C=S;O=A">Size</a></th><th><a href="?C=D;O=A">Description</a></th></tr>
<tr><th colspan="5"><hr></th></tr>
<tr><td valign="top"></td><td><a href="/">Parent Directory</a></td><td></td><td align="right"><a href="nginx_1.24.0-2ubuntu7.5_amd64.deb">nginx_1.24.0-2ubuntu7.5_amd64.deb</a></td><td align="right">2025-08-25 14:28 </td><td align="right">508K</td><td></td><td><a href="packages.gz">packages.gz</a></td><td align="right">2025-11-24 00:43 </td><td align="right">694 </td><td></td></tr>
<tr><th colspan="5"><hr></th></tr>
</table>
<address>Apache/2.4.58 (Ubuntu) Server at localhost Port 8080</address>
</body></html>
root@victor-server:/repo#
```


Paso 11: [Despliegue de la herramienta de automatización] Procedí a instalar el software de gestión de configuraciones ejecutando `sudo apt install -y ansible` en el servidor principal. El sistema descargó los paquetes necesarios, incluyendo las dependencias de Python y las librerías `ansible-core`, preparando el entorno de control desde el cual administraré centralizadamente los nodos remotos del clúster.

```
root@victor-server:/opt# sudo apt install -y ansible
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ansible-core python3-argcomplete python3-dnspython python3-hermes python3-libcloud python3-lockfile
  python3-nimble-auth python3-paslib python3-requests-nimble python3-resolvlib python3-salinas python3-simplejson
  python3-winnn python3-xmldict
Suggested packages:
  ansible-doc python3-trio python3-aiosqlite python3-h2 python3-httpx python3-httpcore python-lockfile-doc
The following NEW packages will be installed:
  ansible ansible-core python3-argcomplete python3-dnspython python3-hermes python3-libcloud python3-lockfile
  python3-nimble-auth python3-paslib python3-requests-nimble python3-resolvlib python3-salinas python3-simplejson
  python3-winnn python3-xmldict
0 upgraded, 15 newly installed, 0 to remove and 4 not upgraded.
Need to get 19.5 MB of archives.
After this operation, 315 MB of additional disk space will be used.
Get:1 http://do.archive.ubuntu.com/ubuntu noble/universe amd64 python3-resolvlib all 1.0.1-1 [25.7 kB]
Get:2 http://do.archive.ubuntu.com/ubuntu noble/main amd64 python3-dnspython all 2.6.1-1ubuntu1 [163 kB]
Get:3 http://do.archive.ubuntu.com/ubuntu noble/universe amd64 ansible-core all 2.16.3-0ubuntu2 [1,280 kB]
Get:4 http://do.archive.ubuntu.com/ubuntu noble/universe amd64 ansible all 9.2.0-0fsg-0ubuntu2 [14.4 kB]
Get:5 http://do.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-argcomplete all 3.1.4-1ubuntu0.1 [33.8 kB]
Get:6 http://do.archive.ubuntu.com/ubuntu noble/universe amd64 python3-hermes all 1.1.14-3.1build1 [21.2 kB]
Get:7 http://do.archive.ubuntu.com/ubuntu noble/universe amd64 python3-lockfile all 1.0.12.2-3 [19.7 kB]
Get:8 http://do.archive.ubuntu.com/ubuntu noble/main amd64 python3-simplejson all 3.19.2-1build1 [54.5 kB]
Get:9 http://do.archive.ubuntu.com/ubuntu noble/universe amd64 python3-libcloud all 3.4.1-3 [71.4 kB]
Get:10 http://do.archive.ubuntu.com/ubuntu noble/universe amd64 python3-nimble-auth all 1.5.0-3 [21.3 kB]
Get:11 http://do.archive.ubuntu.com/ubuntu noble/universe amd64 python3-requests-nimble all 1.1.0-3 [6,308 B]
Get:12 http://do.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-salinas all 0.5-0ubuntu1.1 [163 kB]
Get:13 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 python3-xmldict all 0.13.0-1ubuntu0.21.04.1 [14.3 kB]
Get:14 http://do.archive.ubuntu.com/ubuntu noble/universe amd64 python3-winnn all 0.4.3-2 [31.5 kB]
Get:15 http://do.archive.ubuntu.com/ubuntu noble/main amd64 python3-paslib all 1.7.4-4 [476 kB]
Fetched 19.5 MB in 4s (5,306 kB/s)
Selecting previously unselected package python3-resolvlib.
(Reading database ... 129324 files and directories currently installed.)
Preparing to unpack .../0/python3-resolvlib_1.0.1-1_all.deb ...
Unpacking python3-resolvlib (1.0.1-1) ...
Selecting previously unselected package python3-dnspython.
Preparing to unpack .../01/python3-dnspython_2.6.1-1ubuntu1_all.deb ...
Unpacking python3-dnspython (2.6.1-1ubuntu1) ...
Selecting previously unselected package ansible-core.
Preparing to unpack .../02-ansible-core_2.16.3-0ubuntu2_all.deb ...
Unpacking ansible-core (2.16.3-0ubuntu2) ...
Selecting previously unselected package ansible.
Preparing to unpack .../03-ansible_9.2.0-0fsg-0ubuntu2_all.deb ...
Unpacking ansible (9.2.0-0fsg-0ubuntu2) ...
Setting up python3-resolvlib (1.0.1-1) ...
Setting up python3-dnspython (2.6.1-1ubuntu1) ...
Setting up ansible-core (2.16.3-0ubuntu2) ...
Setting up ansible (9.2.0-0fsg-0ubuntu2) ...
```

Paso 12: [Definición del inventario de servidores] Creé el archivo de inventario para definir los objetivos de la automatización usando `nano hosts.ini`. Dentro del archivo, establecí el grupo `[managed_hosts]` y asigné las direcciones IP y usuarios específicos para cada nodo (`server2` en la IP .120 y `server3` en la IP .106), permitiendo a Ansible identificar correctamente dónde ejecutar las Tareas

```
root@victor-server:~/ansible_project hosts.ini *
GNU nano 7.2
[managed_hosts]
server1 ansible_host=192.168.68.120 ansible_user=server2
server3 ansible_host=192.168.68.106 ansible_user=server3

I

server2@20250664:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 08:00:27:8e:84:67 brd ff:ff:ff:ff:ff:ff
    inet 192.168.68.120/24 metric 100 brd 192.168.68.255 scope global dynamic noprefixroute
        valid_lft 556sec preferred_lft 556sec
    inet6 fe80:a00:27:ff:fe80:a1e4/64 scope link
        valid_lft forever preferred_lft forever
server2@20250664:~$ cat /etc/ufw/user.conf
server2@20250664:~$

server3@20250664:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 08:00:27:8e:84:67 brd ff:ff:ff:ff:ff:ff
    inet 192.168.68.106/24 metric 100 brd 192.168.68.255 scope global dynamic noprefixroute
        valid_lft 556sec preferred_lft 556sec
    inet6 fe80:a00:27:ff:fe80:a1e4/64 scope link
        valid_lft forever preferred_lft forever
server3@20250664:~$ cat /etc/ufw/user.conf
server3@20250664:~$
```

Paso 13: [Intercambio de llaves SSH] Generé un par de claves RSA con ssh-keygen y las distribuí a los nodos para permitir el acceso sin contraseña. Al ejecutar ssh-copy-id, tuve cuidado de corregir un error inicial de usuario y me aseguré de enviar la clave pública al usuario correcto en cada servidor (server2 y server3), garantizando una conexión segura y fluida para la automatización.

```
root@victor-server:~/ansible_project# ssh-keygen -t rsa -b 4096 -N "" -f ~/.ssh/id_rsa
Generating public/private rsa key pair.
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:KqIS8NoYqJvke8QT+7+vn4M0IQylP7S2MedD2oyjE root@victor-server
The key's randomart image is:
+----[RSA 4096]-----+
|
| . .
| o . + + . .
| . O . So + oo +
| o * + . o + o . mo
| + * + . . E o . o .
| = + o . = = . .
| . . . + + + = . .
+----[SHA256]-----+
root@victor-server:~/ansible_project# ssh-copy-id server2@192.168.68.120
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_rsa.pub"
The authenticity of host '192.168.68.120 (192.168.68.120)' can't be established.
ED25519 key fingerprint is SHA256:dQlkoPhXhMaXNATGNVrOCzn4AcHmqKXNRugP5CTcm0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
server2@192.168.68.120's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'server2@192.168.68.120'"
and check to make sure that only the key(s) you wanted were added.

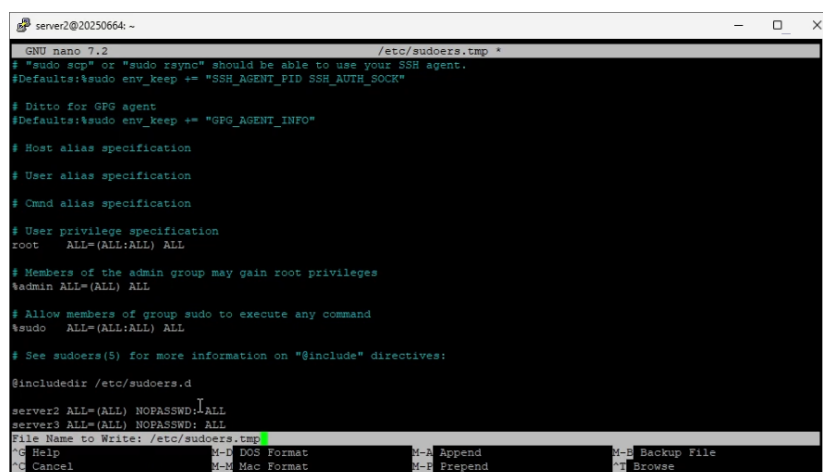
root@victor-server:~/ansible_project# ssh-copy-id server2@192.168.68.106
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_rsa.pub"
The authenticity of host '192.168.68.106 (192.168.68.106)' can't be established.
ED25519 key fingerprint is SHA256:hmVQGIvYj3bUnt5nMoV38U7p9D84S13MB6/z8BCA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
server2@192.168.68.106's password:
Permission denied, please try again.
server2@192.168.68.106's password:
Permission denied, please try again.
server2@192.168.68.106's password:
server2@192.168.68.106's password:
root@victor-server:~/ansible_project# ssh-copy-id server3@192.168.68.106
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
server3@192.168.68.106's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'server3@192.168.68.106'"
and check to make sure that only the key(s) you wanted were added.

root@victor-server:~/ansible_project#
```

Paso 14: [Configuración de privilegios en Nodo 2] Accedí manualmente al servidor server2 para configurar los permisos de superusuario necesarios para Ansible. Edité el archivo de sudoers (similar a usar visudo), agregando la línea server2 ALL=(ALL) NOPASSWD: ALL, lo que permite que el usuario de automatización ejecute comandos administrativos sin interrupciones por solicitud de contraseña.



```
GNU nano 2.2 /etc/sudoers.tmp
# sudo script for "sudo -sync" should be able to use your SSH agent.
#Defaults:sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"

# Ditto for GPG agent
#Defaults:sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin   ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "%include" directives:

@include /etc/sudoers.d

server2  ALL=(ALL) NOPASSWD: ALL
server3  ALL=(ALL) NOPASSWD: ALL
File Name to Write: /etc/sudoers.tmp
[Help] [D] DOS Format [A] Append [B] Backup File
[Ctrl-C] [M] Mac Format [I] Prepend [E] Browse
```

Paso 15: [Configuración de privilegios en Nodo 3] Repetí el procedimiento de seguridad en el servidor server3 para homologar el entorno. Modifiqué el archivo de configuración de sudoers añadiendo la directiva server3 ALL=(ALL) NOPASSWD: ALL, asegurando así que ambos nodos gestionados tengan la capacidad idéntica de elevar privilegios automáticamente cuando el controlador lo requiera.

```
server3@20250664: ~  
GNU nano 7.2 /etc/sudoers.tmp  
# "sudo scp" or "sudo rsync" should be able to use your SSH agent.  
#Defaults:sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"  
  
# Ditto for GPG agent  
#Defaults:sudo env_keep += "GPG_AGENT_INFO"  
  
# Host alias specification  
  
# User alias specification  
  
# Cmnd alias specification  
  
# User privilege specification  
root    ALL=(ALL:ALL) ALL  
  
# Members of the admin group may gain root privileges  
%admin   ALL=(ALL) ALL  
  
# Allow members of group sudo to execute any command  
%sudo    ALL=(ALL:ALL) ALL  
  
# See sudoers(5) for more information on "@include" directives:  
  
@includedir /etc/sudoers.d  
  
server2 ALL=(ALL) NOPASSWD: ALL  
server3 ALL=(ALL) NOPASSWD: ALL  
Save modified buffer?  
Y Yes  
N No      ^C Cancel
```

Paso 16: [Creación del Playbook de Gestión de Usuarios] Desarrollé el script de automatización create_users.yml utilizando nano. Definí una estructura YAML clara para crear los grupos "profs" y "estudiantes", asignar usuarios específicos con contraseñas encriptadas (SHA512) y otorgar privilegios de sudo al grupo de profesores, asegurando una gestión de identidades estandarizada en todos los hosts.

```
root@victor-server: ~/ansible_project  
root@victor-server:~/ansible_project# ansible-playbook -i hosts.ini create_users.yml  
  
PLAY [Crear usuarios en los servidores administrados] *****  
  
TASK [Gathering Facts] *****  
ok: [server2]  
ok: [server3]  
  
TASK [Crear grupos] *****  
changed: [server3] => (item=profs)  
changed: [server2] => (item=profs)  
changed: [server3] => (item=estudiantes)  
changed: [server3] => (item=estudiantes)  
  
TASK [Crear usuarios] *****  
changed: [server3] => (item={'name': 'Joseline', 'group': 'profs'})  
changed: [server2] => (item={'name': 'Joseline', 'group': 'profs'})  
changed: [server3] => (item={'name': 'Margarita', 'group': 'profs'})  
changed: [server2] => (item={'name': 'Margarita', 'group': 'profs'})  
changed: [server3] => (item={'name': 'Danilo', 'group': 'estudiantes'})  
changed: [server2] => (item={'name': 'Danilo', 'group': 'estudiantes'})  
changed: [server3] => (item={'name': 'Hipolito', 'group': 'estudiantes'})  
changed: [server2] => (item={'name': 'Hipolito', 'group': 'estudiantes'})  
  
TASK [Conceder privilegios sudo a profs] *****  
changed: [server2]  
changed: [server3]  
  
PLAY RECAP *****  
server2      : ok=4    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0  
server3      : ok=4    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```


Paso 17: [Ejecución del despliegue de usuarios] Lancé la configuración de identidades ejecutando el comando `ansible-playbook -i hosts.ini create_users.yml`. Supervisé la salida en pantalla, donde Ansible confirmó la conexión exitosa con "Gathering Facts" y reportó el estado "changed" en las tareas de creación de grupos y usuarios, indicando que las modificaciones se aplicaron correctamente en ambos servidores.

```
root@victor-server:~/ansible_project# ansible-playbook -i hosts.ini cron_reboot.yml

PLAY [Configurar reinicio automatico con cron - Victor Santos] *****

TASK [Gathering Facts] *****
ok: [server3]
ok: [server2]

TASK [Programar reinicio diario a las 5:30 a.m.] *****
changed: [server2]
changed: [server3]

TASK [Registrar mensaje en syslog] *****
changed: [server3]
changed: [server2]

TASK [Reiniciar servicio rsyslog] *****
changed: [server2]
changed: [server3]

TASK [Mostrar tareas cron actuales] *****
changed: [server3]
changed: [server2]

TASK [Mostrar tareas cron en pantalla] *****
ok: [server2] => {
  "msg": "$Ansible: Reinicio diario\n30 5 * * * /sbin/shutdown -r now"
}
ok: [server3] => {
  "msg": "$Ansible: Reinicio diario\n30 5 * * * /sbin/shutdown -r now"
}

PLAY RECAP *****
server2      : ok=6    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server3      : ok=6    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

root@victor-server:~/ansible_project#
```

Paso 18: [Programación de tareas de reinicio] Creé un segundo playbook llamado `cron_reboot.yml` para gestionar el mantenimiento programado. En este archivo, configuré una tarea de cron para reiniciar los servidores diariamente a las 5:30 a.m. y añadí una tarea para registrar un mensaje personalizado en el syslog antes del reinicio, asegurando trazabilidad en las operaciones de mantenimiento.

```
root@victor-server:~/ansible_project
GNU nano 7.2                                cron_reboot.yml *
- name: Configurar reinicio automatico con cron - Victor Santos
  hosts: managed_hosts
  become: yes

  tasks:
    - name: Programar reinicio diario a las 5:30 a.m.
      cron:
        name: "Reinicio diario"
        job: "/sbin/shutdown -r now"
        minute: "30"
        hour: "5"

    - name: Registrar mensaje en syslog
      lineinfile:
        path: /etc/rsyslog.d/50-reinicio.conf
        line: "Reinicio iniciado por Victor Santos recién completado"
        create: yes

    - name: Reiniciar servicio rsyslog
      service:
        name: rsyslog
        state: restarted

    - name: Mostrar tareas cron actuales
      command: crontab -l
      register: cron_ver

    - name: Mostrar tareas cron en pantalla
      debug:
        msg: "{{ cron_ver.stdout }}"

File Name to Write: cron_reboot.yml
?G Help          ?M DOS Format    ?A Append       ?B Backup File
?C Cancel        ?d Mac Format    ?o Prepend      ?l Browse
```

Paso 19: [Despliegue de tareas programadas] Ejecuté el playbook de mantenimiento con el comando `ansible-playbook -i hosts.ini cron_reboot.yml` para aplicar las reglas de cron. Verifiqué el resumen "PLAY RECAP", confirmando que todas las tareas se ejecutaron exitosamente (ok=6, changed=4) y validando mediante el módulo de debug que la tarea de reinicio quedó correctamente registrada en el planificador del sistema.

```
root@victor-server:~/ansible_project# ansible-playbook -i hosts.ini cron_reboot.yml

PLAY [Configurar reinicio automatico con cron - Victor Santos] *****

TASK [Gathering Facts] *****
ok: [server3]
ok: [server2]

TASK [Programar reinicio diario a las 5:30 a.m.] *****
changed: [server2]
changed: [server3]

TASK [Registrar mensaje en syslog] *****
changed: [server3]
changed: [server2]

TASK [Reiniciar servicio rsyslog] *****
changed: [server2]
changed: [server3]

TASK [Mostrar tareas cron actuales] *****
changed: [server3]
changed: [server2]

TASK [Mostrar tareas cron en pantalla] *****
ok: [server2] => {
  "msg": "#Ansible: Reinicio diario\n30 5 * * * /sbin/shutdown -r now"
}
ok: [server3] => {
  "msg": "#Ansible: Reinicio diario\n30 5 * * * /sbin/shutdown -r now"
}

PLAY RECAP *****
server2      : ok=6    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server3      : ok=6    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

root@victor-server:~/ansible_project#
```